# GitHub-Dataset.

Data set → QOE for Youtube video streaming.

⇒ subjective
⇒ collected in controlled env
⇒ Based on human Rating.

Key output → MOS (Mean opinion Score)
↳ how user perceived video quality

| MOS | Meaning |
|---|---|
| 5 | Excellent |
| 4 | Good |
| 3 | Fair |
| 2 | Poor |
| 1 | Bad |

Network + video + Device condition

↓

User perceived video quality

# How dataset was collected

The experiment used a **testbed platform** with several tools.

**VLC Player**

Used to play YouTube videos.

VLC also gives **player indicators**, like:

- bitrate
- dropped frames
- resolution
- frame rate

These become **QoA (Quality of Application)** metrics.

**Netem (Network Emulator)**

Used to artificially create network conditions like:

- bandwidth limits
- packet loss
- delay
- jitter

This means they could simulate **bad networks intentionally**.

**iperf**

Used to measure network performance between:

```
Server ↔ testing device
```

This provides **actual QoS measurements**.

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**Python GUI**

Used to:

- show the videos
- collect user feedback
- manage the experiment

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**MySQL Database**

Used to store:

- video metrics
- network metrics
- device info
- user MOS ratings

In total we have 300 samples → each sample is 1 video session under specific condition.

Each sample contains → 29 QoE influence factors + MOS

∴ in total we have 30 clms

The columns are very organised here

## A) Network QoS Parameters (2–5)

These represent **network conditions**.

Columns:

| Column | Meaning |
|---|---|
| QoS_bandwidth | network bandwidth |
| QoS_packet-loss | packet loss rate |
| QoS_delay | network delay |
| QoS_jitter | delay variation |

These simulate **real network problems**.

## B) Video Content Features (6–10)

These describe the **video itself**.

| Column | Meaning |
|---|---|
| V_id-video | video ID |
| V_content | video category |
| V_norm-bitrate | normalized bitrate |
| V_complexity | video complexity score |
| V_complexity-class | complexity category |

These are important because:

Different videos behave differently under network issues.

## C) VLC Player Indicators (QoA) (11–16)

These are **metrics measured by VLC player**.

| Column | Meaning |
|---|---|
| QoA_VLCresolution | video resolution |
| QoA_VLCbitrate | bitrate used |
| QoA_VLCframerate | frames per second |
| QoA_VLCdropped | dropped frames |
| QoA_VLCaudiorate | audio bitrate |
| QoA_VLCaudioloss | audio loss |

## D) Device Characteristics (QoD) (17–29)

These describe the **device hardware**.

Examples:

| Column | Meaning |
|---|---|
| QoD_cpu_mhz | CPU speed |
| QoD_cpu_core_nbr | number of cores |
| QoD_ram_size | RAM size |
| QoD_screen_resolution | screen resolution |
| QoD_screen_dimension | screen size |
| QoD_ram_clock_speed | RAM speed |

and finally we have MOS → ground truth of QoE score

why this dataset is valuable?

```
QoS → network conditions
QoA → application playback behavior
QoD → device capability
QoU → user perception
```

as it includes 4 layers of QoE influence factor.

# Results from EDA

The packet loss parameter remained constant at 0% across all experiments. Therefore, it does not contribute to QoE variation in this dataset and was excluded from correlation analysis.

```
id                  0.890642
QoS_bandwidth       0.839049
```

**Problem**

`id` should **not correlate with MOS**.

Why?

Because `id` is just a row identifier.

**Why it happened**

Rows were probably **sorted by quality level**.

therefore we dropped the id clm

Top features affecting QoE:

```
code

QoS_bandwidth        0.839
QoA_audio-rate       0.778
QoA_resolution       0.744
QoA_frame_rate       0.719
QoA_bitrate          0.719
```

**Interpretation**

This means:

```
code

Higher bandwidth → higher MOS
Higher video resolution → higher MOS
Higher frame rate → higher MOS
Higher bitrate → higher MOS
```

| MOS | Bandwidth Range |
|---|---|
| 1 | ~1000–1500 |
| 2 | ~1000–1600 |
| 3 | ~1500–2000 |
| 4 | ~2500–3500 |
| 5 | ~3000–4600 |

| Resolution | MOS |
|---|---|
| 144p | mostly 1–2 |
| 240p | mostly 2–3 |
| 360p | around 3–4 |
| 480p | around 4 |
| 720p | around 4–5 |
| 1080p | mostly 5 |